VLSI Design Internship – Task 2 Report

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Branch: EC 3rd Year

Topic: Verilog HDL and RTL Design of Basic Combinational Circuits

Tools Used: EDA Playground

Aim

To design, implement, and verify basic combinational circuits using Verilog HDL, including logic gates, a Half Adder, and a Full Adder, and analyze their waveforms.

Objectives

- Understand Verilog HDL syntax and module structure.
- Design combinational circuits using continuous assignment.
- Implement logic gates and adder circuits.
- Write testbenches and verify outputs.
- Analyze waveform outputs.

Theory

Verilog HDL is used to describe digital systems. In RTL design, circuits are described in terms of data flow and logical operations. Combinational circuits depend only on present inputs. Testbenches apply inputs and verify outputs. Waveforms help visualize signal changes over time.

In Verilog, a digital system is modeled using **modules**, which act as building blocks of hardware design. Each module consists of input and output ports, internal signals, and behavioral or structural descriptions of the circuit. The behavior of combinational circuits is commonly described using **continuous assignment statements (assign)**, where outputs are expressed as logical functions of inputs.

Verilog supports various **data types** such as wire and reg. Wires are used to represent connections between components, while registers are used to store values in procedural blocks. For combinational logic design, wires are mainly used along with continuous assignments.

Logical operations in Verilog are performed using operators such as:

- AND (&)
- OR (|)
- NOT (~)
- XOR (^)

These operators directly map to hardware logic gates, making Verilog suitable for describing digital circuits at a low level.

A **testbench** is a separate module used only for simulation purposes. It generates different combinations of input signals and observes the corresponding outputs from the design under test (DUT). Testbenches are not synthesized into hardware but are essential for verifying correctness.

Simulation plays a crucial role in digital design. It allows designers to check whether the circuit behaves as expected before actual hardware implementation. During simulation, a **VCD (Value Change Dump) file** is generated, which stores signal transitions over time. This file is then viewed using waveform viewers to analyze timing and logic behavior.

Waveform analysis helps in:

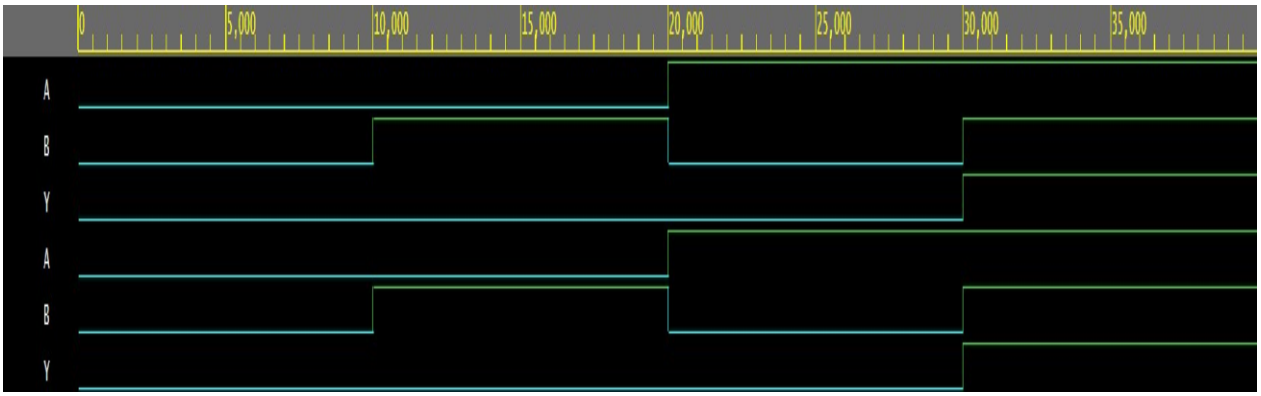
- Verifying correctness of output
- Observing signal transitions
- Detecting errors or glitches
- Understanding timing relationships

Thus, Verilog HDL combined with simulation and waveform analysis provides an efficient way to design, test, and validate digital circuits before physical realization.

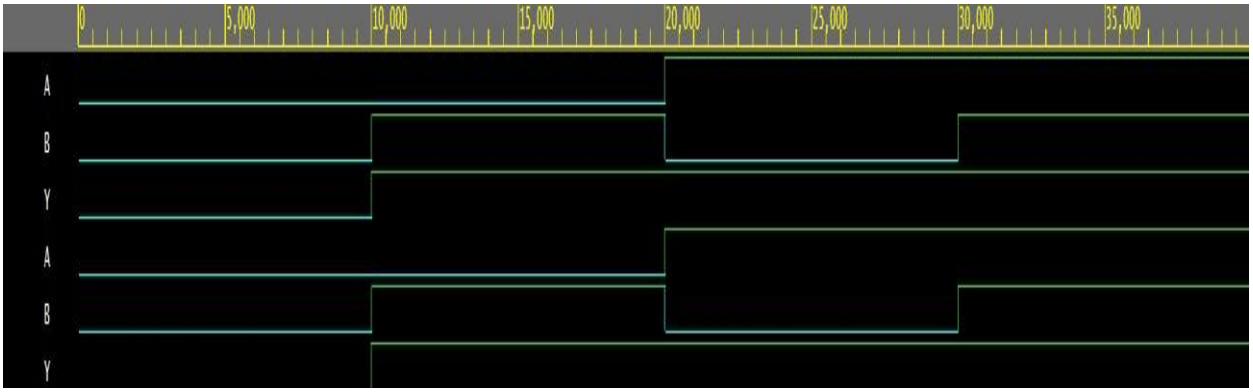
Logic Gates

- AND: Y = A · B
- OR: Y = A + B
- NOT: Y = ¬A
- NAND: Y = ¬(A · B)
- NOR: Y = ¬(A + B)
- XOR: Y = A □ B

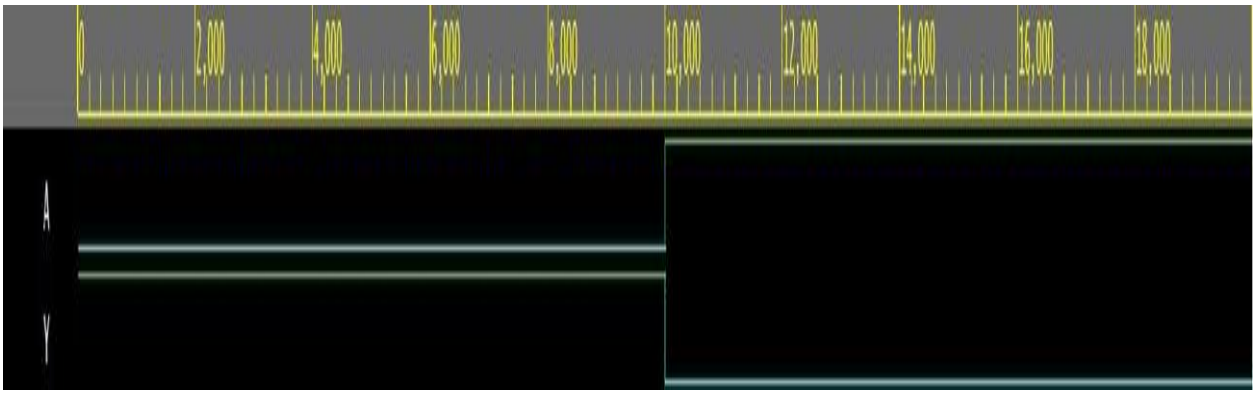
Waveforms



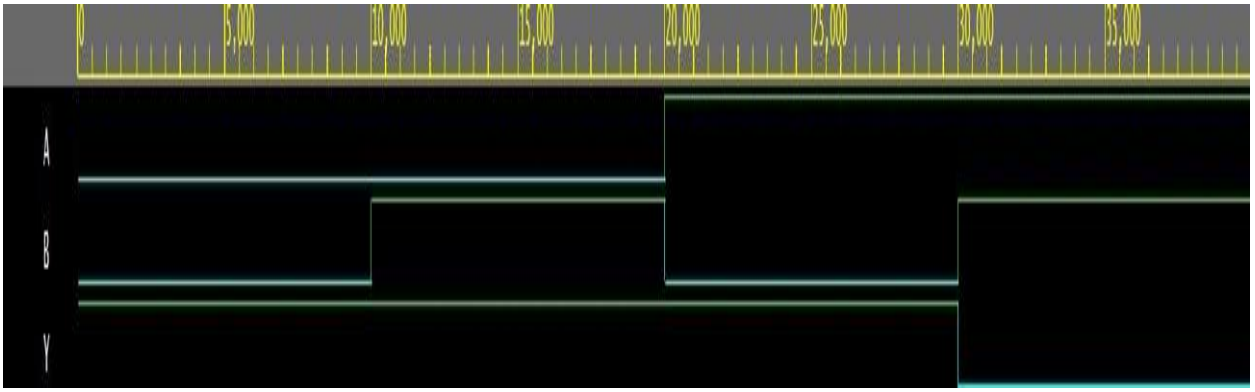
AND Gate



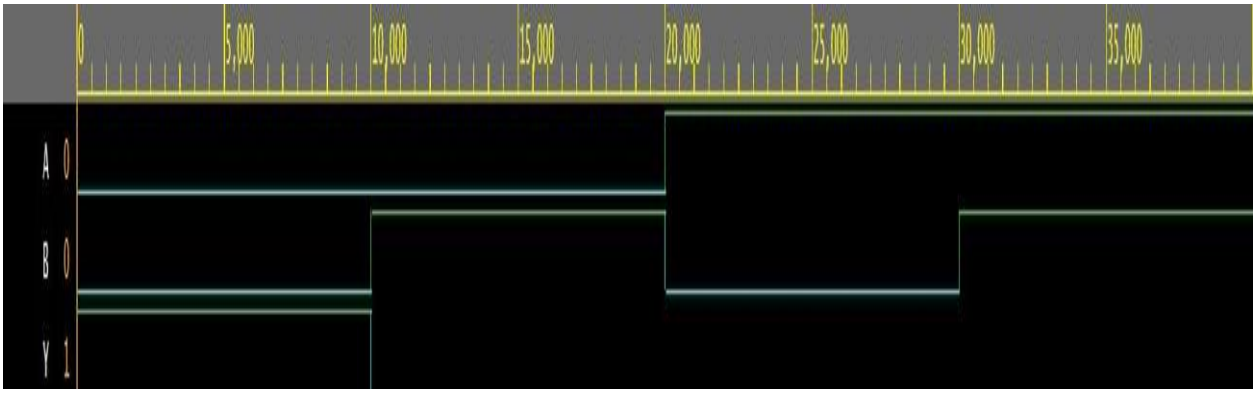
OR Gate



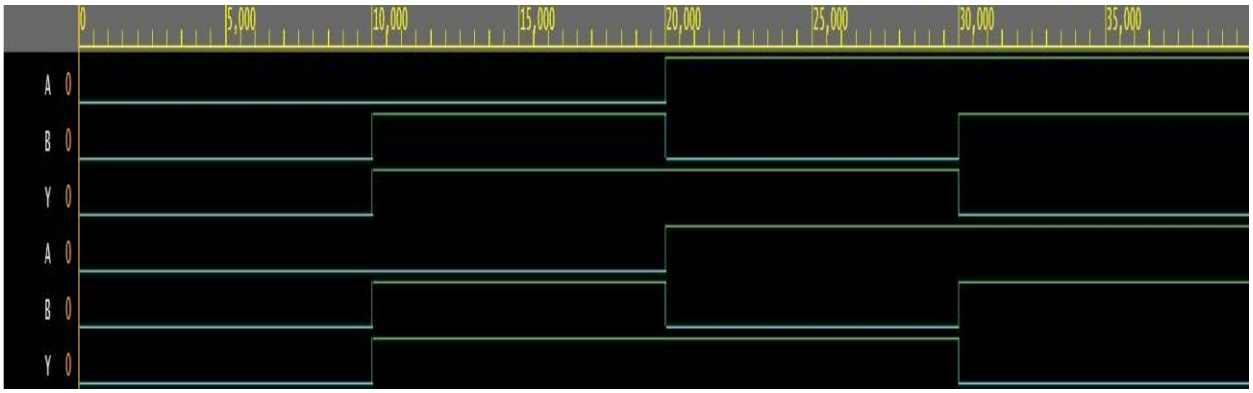
NOT Gate



NAND Gate



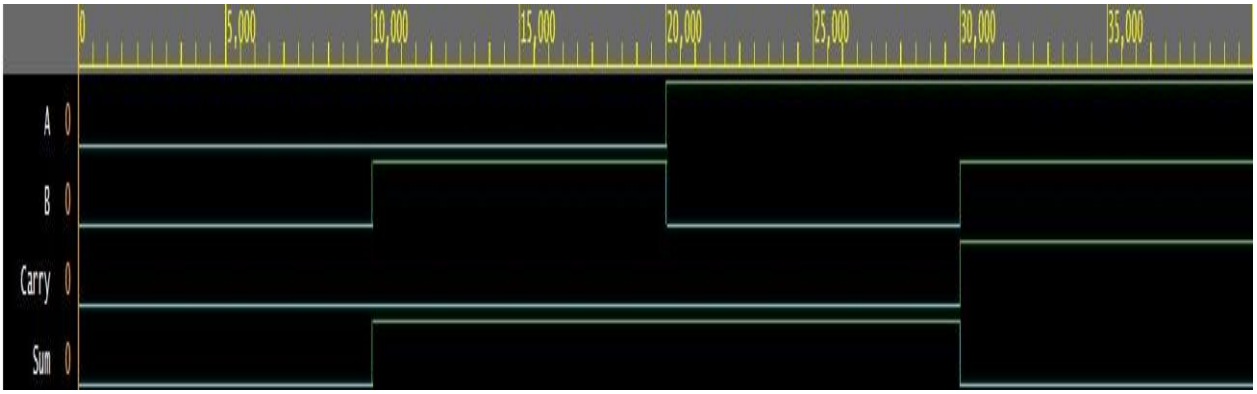
NOR Gate



XOR Gate

Half Adder

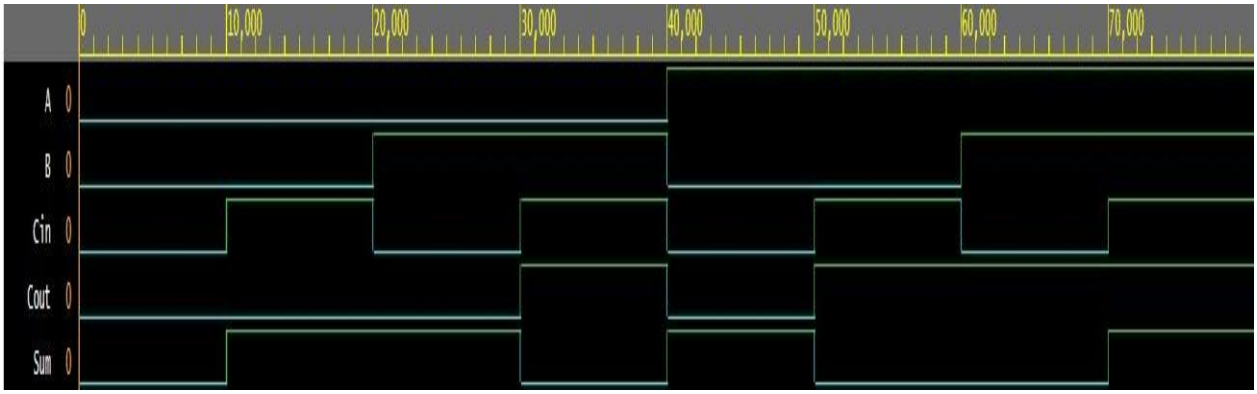
Adds two bits. Sum = A □ B, Carry = A · B.



Half Adder Waveform

Full Adder

Adds three bits. Sum = A □ B □ Cin, Cout = AB + BCin + ACin.



Full Adder Waveform

Conclusion

All circuits were successfully implemented and verified using EDA Playground. Outputs matched expected results.

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